

02 — N-ATLaS integration evidence

PauseAm | Ask before you pay. Status: BLOCKED for qualifying PS2 submission. Actual code and metadata are documented here. Actual text loading succeeded, but no guidance inference has passed the response contract. Four-language voice remains blocked.

Architecture

Browser consent and microphone → WAV conversion → same-origin /api/asr → configured HTTPS ASR host → pinned official model → identity validation → editable transcript → explicit confirmation → /api/answer → current source library.

When enabled, the pinned N-ATLaS text service selects one source card or abstains. The browser renders source-linked wording, never model-supplied links, contacts or free-form instructions. Urgent bank-contact actions do not wait for a model.

Code evidence

lib/models.ts records official identities and immutable revisions. lib/asr.ts and app/api/asr/route.ts enforce consent, audio bounds and provenance. components/voice-input.tsx handles consent, recording, playback, upload and cancellation. components/journey.tsx requires transcript correction/confirmation. asr/app.py uses the documented official Transformers pipeline. text_service/app.py uses AutoTokenizer, AutoModelForCausalLM and apply_chat_template. lib/text-model.ts validates constrained card selection. lib/official-api.ts fails closed until the legitimate contract is supplied.

Redacted configuration

ASR_ENABLED=false, TEXT_ENABLED=false, KB_ENABLED=true. ASR_ENDPOINT, ASR_YO_ENDPOINT, ASR_HA_ENDPOINT, ASR_IG_ENDPOINT and TEXT_ENDPOINT remain unset in the public app until their genuine end-to-end checks pass. The text host exists at <https://copper-squirrel.vm.scrtlabs.com/text/guide>; its first guidance response failed the contract. Tokens belong only in host secret settings. HF_TOKEN belongs only on model hosts. MODEL_LANGUAGE and MODEL_REVISION explicitly select the official weights. No credentials appear in this document.

Genuine request evidence

Public Hugging Face metadata returned the five pinned revisions and file lists. Anonymous configuration probes for all four pinned ASR models returned HTTP 401 on 2 October. Later authenticated probes returned HTTP 403 for all four; the provider reported that the token's account was not authorized for these models. Local HTTP checks exercised retrieval and unavailable inference; they did not perform speech recognition. The authenticated text-host health check returned HTTP 200 with the exact model revision, verified-bucket provenance and bfloat16 dtype. This confirms genuine weight loading. The first actual guidance request returned HTTP 422 after 30,453 ms: generated output failed the strict contract. The non-sensitive engineering input is synthetic; model execution and its failure are real. No participant interaction or successful guidance is inferred from it. See submission/evidence/text-inference-first-attempt-2026-10-02.json and submission/evidence/asr-access-2026-10-02.json and submission/evidence/asr-authenticated-access-2026-10-02.json. The parser update passed 16 unit tests and built in run 36956992856; the provider subsequently reported an update error. Recovery and live retest remain necessary. No language screenshots exist.

After actual consented calls, capture only timestamp, build, language, model/revision, request ID, response status, duration, bytes and identity-check result. Do not log audio, text, secrets or full headers. Separately consent to a non-sensitive scripted example for screenshots; never invent a transcript to fill the evidence gap.

Claim decisions

All four ASR adapters: implemented, live inference unverified. N-ATLaS text adapter: genuine weights loaded; actual request failed safely; live guidance not passed. Official API: no documented contract or successful request. Fine-tuning: no training run or held-out comparison. No capability checkbox is justified by the current evidence. Organiser confirmation of the official-ASR-service requirement is pending. See the following access register for exact versions, licence and compute findings.

User-operated deployment handoff

The updated bucket root fully matches the official Igbo model's 13 required files, including the entire binary weight. The other three ASR weights do not match; text configuration files have been overwritten by ASR files. Five isolated folders are required for reproducible deployment. These byte checks are not speech output. The final images built successfully in run 36959714636 from source 836fe48; their actual digests and anonymous access are recorded in model-container-build-2026-10-02.json. The decoder now constrains actual model token choices to one trusted card or abstention, then independently validates JSON. Eighteen Python checks passed. Final-image runtime inference remains unverified until deployment and real tests. See deployment/secretvm/README.md, pauseam-compose.yml and scripts/verify-model-host.mjs.

PauseAm — model access findings

Checked 1 October 2026; deployment findings updated 2 October. No successful guidance response has yet passed the output contract. Public model cards and file metadata were read. Gated configuration/weight files could not be read: a direct official Yoruba config request returned HTTP 401. At the initial inspection no Hugging Face token was available. A later private token was tested on 2 October; each ASR configuration request returned HTTP 403, with the provider reporting GatedRepo access restrictions. See ../evidence/asr-authenticated-access-2026-10-02.json. File-list inspection is not weight inspection and is not proof of model integration.

Official repositories and pinned versions

- Text: <https://huggingface.co/NCAIR1/N-ATLaS> — e294476928aca9030e924ca27bb8e085e8581273 - English: <https://huggingface.co/NCAIR1/NigerianAccentedEnglish> — 3c52c6e6c9ec508014a7b9db6a42b503b8930dff - Yoruba: <https://huggingface.co/NCAIR1/Yoruba-ASR> — d1ae7b8b79c2ccd547d8761effe5057433f3fc7f - Hausa: <https://huggingface.co/NCAIR1/Hausa-ASR> — e635b9eda29060c6114c8f4d8b2d903f5c83a44a - Igbo: <https://huggingface.co/NCAIR1/Igbo-ASR> — 180732299d5cba3dc8b289260ac84b7838bb3954

All five metadata records report automatic gating. Account holders must accept publisher conditions. The four ASR cards document a Transformers speech-recognition pipeline with 16 kHz audio. The text card documents AutoTokenizer, AutoModelForCausalLM and apply_chat_template. Our services follow those entry points, add pinned revisions and reject identity mismatches.

ASR file lists contain pytorch_model.bin, not Safetensors. The previous server incorrectly required Safetensors; it now uses PyTorch weights-only loading with torch 2.6.0 and remote code disabled. This change has not yet been exercised against downloaded weights. Text files list four Safetensors shards. No optimiser or training-state pickle files are needed or loaded by the inference services.

Service versus weights versus official API

The NAIC PS2 rule explicitly calls for the official N-ATLaS ASR service: <https://ncair.nitda.gov.ng/naic/> The page lists API credentials as shortlisted-team support. No verified official service URL or API contract was found in the inspected official pages. Self-hosted official weights may or may not satisfy PS2; organiser confirmation is pending. The prepared organiser query has not been sent.

The web adapters point only at explicitly configured HTTPS hosts. Such a host would be a team-operated inference service, not proof of official API use. lib/official-api.ts always fails closed until a documented contract is available.

Capability claim register

- English ASR: adapter implemented; live result unverified. - Yoruba ASR: adapter implemented; live result and fluent review unverified. - Hausa ASR: adapter implemented; live result and fluent review unverified. - Igbo ASR: adapter implemented; live result and fluent review unverified. - Text N-ATLaS: real pinned weights loaded on the team host; first actual inference failed the strict contract after 30,453 ms. Guidance was withheld. Fix and retest pending. - Official API: no successful request; do not check this capability. - Fine-tuning: not performed; do not check this capability. - End-to-end four-language guidance: blocked; English interface/source wording only.

The bucket was subsequently updated. The complete root Igbo copy matched all 13 required file hashes, including its full 967 MB weight. Other ASR weight hashes did not match. Text configuration/tokenizer files were replaced by ASR files; text weight shards remain but were not fully re-downloaded in this audit. See `../evidence/bucket-audit-2026-10-02.json`. Keep five separate model directories. Igbo file verification is not actual ASR inference or four-language validation. Deployment is now user-operated; instructions and pinned Compose are in `../deployment/secretvm/README.md`.

Resources actually inspected

Windows host: 8 logical processors; approximately 31.8 GiB RAM, 7.9 GiB free at inspection; NVIDIA Quadro P520 with 4,096 MiB VRAM. Docker CLI is installed but its Linux engine was not running. Node dependencies are installed. The bundled Python, GitHub CLI, FFmpeg and Poppler are available. GitHub CLI identifies thepopeblack-byte. A paid Google Cloud SecretVM was subsequently provisioned on 2 October: eight vCPUs, 32 GB RAM, 160 GB disk, \$0.52/hour running and \$0.052/hour stopped. Its HTTPS certificate was verified. It is a team host, not the official N-ATLaS API.

Four ASR models may be run on separate small hosts or sequentially on suitable compute after measurement. The text model needs substantially more memory than available GPU capacity; no local full-model training claim is plausible. Choose a host based on measured load, not an invented cost/performance number.

Licence and attribution

Each model card publishes terms with a cap of 1,000 active end users in a rolling 30-day period. Separate commercial licensing is required before exceeding it, through Awarri Technologies in partnership with FMCIDE. Contact details appear in the official model cards: `datasupport@awarri.com` and `ncair@nitda.gov.ng`. Obtain a written licence decision before scaling; do not assume a software licence for this repository grants model rights. Required model attribution:

"N-ATLaS is an initiative of the Federal Ministry of Communications, Innovation and Digital Economy, and powered by Awarri Technologies."

No model is renamed by PauseAm. Keep model attribution with all public use. Before enabling public inference, implement and verify cross-host rolling user accounting or restrict the pilot to a tracked roster below the cap. A browser cookie is not reliable person counting. Rate limits alone do not enforce this cap. The implemented persistent shared quota allows at most 950 inference reservations per rolling 30 days, counting failures and recording timestamps only. All five services must share the same database; never reset or fork it to bypass the ceiling. Current public inference remains disabled; the proposed pilot pool is 200+, not an approved unlimited model audience.